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Multi-tracker meta-search for qBittorrent â€” self-hosted, containerised, private-tracker-aware.

[Quick start](#) Â· [Tokens & keys](#) Â· [Documentation](#) Â· [Architecture](#) Â· [Testing](#) Â· [Contributing](#)

python tests	5437 collected	frontend tests	371 collected	plugins	43	merge service	FastAPI :7187
ci	manual (./ci.sh, no auto-trigger)	pre-build invariants	46	challenges	38		
scanners	snyk sonar bandit ruff semgrep trivy gitleaks pip-audit					license	Apache 2.0

Features

- **Merge Search Service** â€” FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** â€” result_found events arrive as each tracker completes, no blocking.
- **Private-tracker bridge** â€” Authenticated downloads via webui-bridge.py for RuTracker, Kinozal, NNM-Club, IPTorrents.
- **Freeleech-only IPTorrents** â€” Automation never costs ratio (see [constitution VIII](#)).
- **Opt-in quality stack** â€” SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
- **Opt-in observability** â€” Prometheus + Grafana dashboards behind the same profile system.
- **Hook system** â€” Register scripts via POST /api/v1/hooks for search / download events.
- **Angular 21 SPA dashboard** â€” dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
- **PWA-ready** â€” favicon + launcher icons + web manifest ship from frontend/public/.

- **Container-first** – rootless Podman or Docker, single `./start.sh` boot.
 - **Resource-pressure early-warning** – [challenges/scripts/resource_pressure_signature_challenge.sh](#) checks 5 independently-falsifiable host-pressure signatures (runaway RSS, thread-count vs `ulimit -u`, container EAGAIN cascades, swap pressure, cgroup memory pressure) so a forced-logout precursor is caught *before* `user@1000.service` gets SIGKILLED, not after. Wired into `scripts/pre_build_verification.sh` invariant 25 and a standing `boba-resource-pressure-check.timer` systemd-user unit. See [incident writeup](#).
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Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent
```

Optional: configure tracker credentials + tokens

```
cp .env.example .env
```

```
$EDITOR .env # see "Tokens & API keys" below
```

```
./setup.sh # one-time
```

```
./start.sh # Angular build + containers + plugin sync
```

Then visit:

URL	Purpose	Auth
http://localhost:7187/	Merge-search dashboard (SPA)	–
http://localhost:7187/docs	FastAPI Swagger UI	–
http://localhost:7187/openapi.json	OpenAPI schema	–
http://localhost:7186/	qBittorrent WebUI proxy	admin / admin
http://localhost:7185/	qBittorrent internal (container)	–
http://localhost:7188/	WebUI bridge (private-tracker downloads)	–

Tokens & API keys

– **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS_AND_KEYS.md](#) – which variable is mandatory, which is optional, and **where to register** for each one.

Tracker credentials “ env vars AND cookies files (auto-loaded)

Boba supports **two mechanisms** for tracker auth, and both are respected together:

1. **Env vars in .env** “ <TRACKER>_USERNAME / <TRACKER>_PASSWORD / <TRACKER>_COOKIES.
2. **Cookies files in ~/Downloads/** “ canonical naming cookies_<tracker>.txt (lowercase), Netscape TSV format exported from a logged-in browser session.

The loader scripts/load-tracker-cookies.sh auto-invokes before every boba-svc up, boba-svc restart, install.sh Stage 6, and start.sh boot “ so refreshing a session is a one-step *browser re-export* “ restart. Source directory defaults to \$HOME/Downloads, overridable via TRACKER_COOKIE_DIR or --dir <path>.

See the [Tracker Credentials Manual](#) for the per-tracker matrix, browser-export step-by-step, precedence rules, and the security posture (Â§11.4.10.A leak audit, cookie-value-never-logged). Common questions: [FAQ](#).

Downloads and directories the system creates are owned by the account that started it “ no chown step after every download. See the [File Ownership Guide](#) for why that requires PUID=0 under rootless Podman, how to verify it on your own machine, and how to repair content created before the fix.

At a glance

Category	Mandatory	Optional	Documentation
qBittorrent WebUI	“	WEBUI_USERNAME, WEBUI_PASSWORD	Â§1
Private trackers	per-tracker	“	Â§2
Public tracker uplift	“	JACKETT_API_KEY, “	Â§3
Metadata enrichment	“	TMDB_API_KEY, TVDB_API_KEY, “	Â§4
Security scanning	“	SNYK_TOKEN, SONAR_TOKEN, “	Â§5
Observability	“	GRAFANA_USER, GRAFANA_PASSWORD	Â§6
Orchestrator tuning	“	ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, “	Â§7

Where to register (fast links)

Provider	Signup	Notes
RuTracker	https://rutracker.org/forum/register.php	Username + password
Kinozal	https://kinozal.tv/signup.php	May require invite
NNM-Club	https://nnmclub.to/forum/ucp.php?mode=register	Cookie-based auth
IPTorrents	https://iptorrents.com/	Invite-only; freeleech enforced
Jackett	https://github.com/Jackett/Jackett	Self-hosted indexer
TMDb	https://www.themoviedb.org/signup	v3 auth API key
TVDb	https://thetvdb.com/api-information	Subscription API
MusicBrainz	https://musicbrainz.org/doc/MusicBrainz_API	Free â€” needs UA only
Snyk	https://app.snyk.io/	Account â†’ Auth Token
SonarCloud	https://sonarcloud.io/account/security/	Or run the compose SonarQube
Gitleaks	https://gitleaks.io/	Optional commercial license

Documentation

Everything the platform offers, indexed:

Tracked-Items + Status Documents

The mandated entry point (Â§11.4.57 / Â§11.4.212) for every tracker and status document in this repository â€” Last modified and Revision are sourced verbatim from each documentâ€™s own revision header (Â§11.4.44); a â€” means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated).

Document	Last modified	Revision	Markdown	HTML
Issues	2026-08-18T19:17:52Z	10	Markdown	HTML
Issues_Summary	â€”	â€”	Markdown	HTML
Fixed	2026-08-18T20:55:14Z	20	Markdown	HTML
Fixed_Summary	â€”	â€”	Markdown	HTML
CONTINUATION	2026-08-18T21:10:20Z	24	Markdown	HTML
PORTING-FROM-LAVA	2026-07-01T16:18:43Z	1	Markdown	

Document	Last modified	Revision	Markdown	HTML
REMAINING_WORK_PLAN	2026-08-07T19:10:54Z	2	Markdown	HTML
GOVERNANCE_AUDIT_2026-08-07	2026-08-07T19:10:54Z	1	Markdown	HTML
browser_extension/Status	2026-06-13T13:10:00Z	15	Markdown	HTML
browser_extension/ Status_Summary	2026-06-10T20:05:00Z	3	Markdown	HTML
browser_extension/ RELEASE_READINESS	2026-06-13T13:10:00Z	4	Markdown	HTML
codegraph/Status	2026-08-18T13:33:41Z	2	Markdown	HTML
codegraph/Status_Summary	2026-08-20T12:27:23Z	1	Markdown	HTML
COMPLETION_STATUS	2026-06-06T00:00:00Z	1	Markdown	HTML
RELEASE_READINESS_20260616	2026-06-16T10:30:00Z	1	Markdown	HTML
features/Status	2026-08-18T13:33:41Z	8	Markdown	HTML
features/Status_Summary	2026-06-16T23:30:00Z	7	Markdown	HTML
QA_DISCOVERY_LEDGER	2026-08-18T22:10:00Z	9	Markdown	HTML

Getting started & operation

- [docs/USER_MANUAL.md](#) â€” end-user walkthrough
- [docs/TOKENS_AND_KEYS.md](#) â€” **â€” credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN_SYSTEM.md](#) â€” runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) â€” the 43 plugin engines
- [docs/PLUGIN_TROUBLESHOOTING.md](#) â€” what to check when a plugin breaks

Architecture & subsystems

- [docs/architecture/](#) â€” 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA_MODEL.md](#) â€” Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) â€” asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) â€” Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) â€” benchmark + load + stress test layout
- [docs/api/openapi.json](#) â€” frozen OpenAPI spec (diffed in CI)

Quality & security

- [docs/SECURITY.md](#) â€” threat model + credential storage

- [docs/SCANNING.md](#) â€” Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY_STACK.md](#) â€” the opt-in docker-compose.quality.yml stack

Incidents & QA evidence

Per Â§11.4.238, automated HelixQA coverage is the discovery layer â€” manual/operator/agent-found defects are confirmation-only, and every out-of-band finding gets a coverage-escape-audit entry here.

- [docs/QA_DISCOVERY_LEDGER.md](#) â€” the discovery-channel ledger + coverage-escape schema (also indexed above in Tracked-Items)
- [docs/incidents/](#) â€” host-safety + CONST-033 forensic investigations, including the latest [2026-08-18 2nd perceived forced-logout](#) triage (root cause: partial, Â§11.4.6 UNCONFIRMED: boundary honestly stated; preventive gate: the resource-pressure signature challenge above)
- [docs/qa/](#) â€” per-item captured machine evidence (Â§11.4.5 / Â§11.4.69 / Â§11.4.83), one directory per workable item / task run â€” e.g.Â [docs/qa/BOB-116/](#)

Manual QA

Per Â§11.4.185, automated gates are necessary but not sufficient â€” every release deliverable requires a final human confirmation pass. Per-session manual-QA checklists live under [docs/manual-qa/](#), one file per session, in the Â§11.4.153 per-feature Status/video-confirmation format.

- [docs/manual-qa/2026-08-18-session-manual-qa-checklist.md](#) â€” 12-feature checklist covering the boba-stack reachability, cross-tracker search/dedup, download flow, BOB-112 /healthz DDoS mitigation, constitution v68 landing, resource-pressure timer, CONST-033 challenge, and pre-build gate for the 457cca4â†’63f3f88 session range

Testing

- [docs/TESTING.md](#) â€” catalogue of every test type (30 rows)
- [docs/COVERAGE_BASELINE.md](#) â€” per-module coverage gates
- [tests/README.md](#) â€” layout of the tests/ tree

Development

- [CONTRIBUTING.md](#) â€” contribution workflow
- [CLAUDE.md](#) â€” Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) â€” runtime development guidance
- [.specify/memory/constitution.md](#) â€” **binding architectural contract (v1.1.0)**
- [docs/scripts/](#) â€” per-script user guides (Â§11.4.18), one <name>.md per scripts/*.sh / challenges/scripts/*.sh â€” e.g.Â [commit-push-all.md](#), [capture-workable-items-db-delta.md](#), [regenerate-continuation-exports.md](#)

Courses (self-paced, Asciiinema)

- [courses/01-operator/](#) â€” Your first search
- [courses/02-plugin-author/](#) â€” Authoring a nova3 plugin
- [courses/03-contributor/](#) â€” TDD + rebuild-reboot deep dive
- [courses/04-security-ops/](#) â€” Threat model + scanner bundle

Release / CI

- [CHANGELOG.md](#)
- [releases/README.md](#) â€” release artefacts layout
- [docs/OUT_OF_SANDBOX.md](#) â€” items requiring external credentials (HelixQA / OpenCode / submodule orgs)

Architecture

```
graph LR
    subgraph "http://:7186"
        qbittorrent-proxy
        python3.12-alpine
    end
    subgraph "http://:7187"
        Merge-search-service
        Angular-SPA-FastAPI
    end
    subgraph "http://:7188"
        webui-bridge
    end

    qbittorrent-proxy --> Merge-search-service
    python3.12-alpine --> Merge-search-service
    Merge-search-service --> Angular-SPA-FastAPI
    webui-bridge --> Merge-search-service
```

Download proxy ([:7186](#))

Merge search service ([:7187](#))

Angular SPA + FastAPI

[/api/v1/search](#) + SSE

[/api/v1/bridge/health](#)

[/api/v1/auth/...](#)

webui-bridge.py (host process) â€” private-tracker b

Opt-in (docker-compose.quality.yml):

http://:9000	SonarQube	profile: quality
http://:9090	Prometheus	profile: observability
http://:3000	Grafana	profile: observability
(run-once)	Snyk / Semgrep / Trivy / Gitleaks	profile: run-once

Full Mermaid renders in [docs/architecture/](#).

Testing

```
# Non-live-HTTP suites â€” run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks â€” dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security â€” hits the live merge service, needs stack up
```



```
python3 -m pytest tests/security/ --no-cov -q
```

```
# Integration â€” same
```

```
python3 -m pytest tests/integration/ --no-cov -q -m "not requires_credentials"
```

```
# Frontend
```

```
npx --prefix frontend ng test --watch=false
```

```
# Full scanner sweep (non-interactive; skips scanners with missing tokens)  
./scripts/scan.sh --all
```

```
# Pre-build invariant sweep â€” 25 checks (workable-items integrity, docs-  
# sync, resource-pressure signature, mutation-residue, Â§11.4.238 discover  
# ledger freshness, ...) run before every build  
./scripts/pre_build_verification.sh
```

```
# HelixQA Challenges â€” 31 scripts under challenges/scripts/, e.g.:  
./challenges/scripts/resource_pressure_signature_challenge.sh  
./challenges/scripts/ddos_resilience_challenge.sh --healthz
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh          # all targets, all channels  
./scripts/build-releases.sh frontend # one target  
./scripts/build-releases.sh --channel release
```

Each artefact ships with SHA256SUMS + BUILD_INFO.json.

Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest â€” 4474 tests collected, see docs/TESTING.md)
 - Frontend Vitest (ng test â€” 371 tests collected, see docs/TESTING.md)
 - Ruff + bandit + shellcheck (via scripts/scan.sh)
-

License

Apache 2.0 â€” see [LICENSE](#).